

From Theory to Desk: Quant Tricks That Actually Matter

A Practical Guide to Greeks, Vol Surface/Smiles, YC in IR, Time-Series Models, Numerical Methods & Monte Carlo and Risk/XVA

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"Look-Ahead Bias is Time Travel"**Memory:** If your backtest Sharpe > 3, you probably time-traveled. **Fix:** Always lag data by 1 period.

Data Snooping:

Test 100 strategies, pick best 1 $\implies$ overfit noise

Fix: Bonferroni correction, out-of-sample testing.

Overfitting:

Too many parameters relative to observations

Rule: Need 20-30 observations per parameter.

Walk-Forward Validation:

Train on $[0, t]$, test on $[t + 1, t + \Delta]$, roll forward

Why: Mimics live trading. **Never:** Train on all data, test on subset.

Backtesting Checklist"Assume your future self is dumber than your past self"Use walk-forward, lag all features, keep 30% OOS"I start with 5 years train, 1 year test, roll quarterly. If Sharpe drops >50% OOS, I overfit."

Part 5: Interview Toolkit

13. "Which Model When" Cheat Table

Use Case	Model	Why	Watch Out
Daily PnL for VaR	GARCH(1,1) or EWMA	Captures vol clustering	Overfits if long memory
Intraday tick data	Hawkes process	Excitation effects	Execution costs dominate
Pair trading	Engle-Granger cointegration	Mean-reverting spread	Regime breaks
Yield curve	3-factor VAR or Nelson-Siegel	Curve dynamics	Dimensionality
Macro forecasting	Regime-switching VAR	Structural breaks	Slow to detect
Vol surface modeling	GARCH with exogenous vars	Vol-of-vol	Skew not captured
Credit spreads	AR(1) + jumps	Spikes	Jump risk hidden
High-frequency arb	ARIMA(0,0,0) (white noise)	No predictability	Latency

14. Q&A Templates: Interview Soundbites

1. How would you model daily PnL of an options book for VaR? You have long Gamma, short Vega. **Answer:** "Two-step approach:

1. **Factor model:** $PnL_t = \Delta \cdot r_t + 0.5\Gamma \cdot r_t^2 + \mathcal{V} \cdot \Delta\sigma_t + \epsilon_t$. Estimate $\Delta, \Gamma, \mathcal{V}$ from risk system.
2. **Vol model:** Fit GARCH(1,1) to r_t and σ_t to capture clustering. Use t-distribution for residuals (fat tails). Run 10-day VaR via Monte Carlo: simulate σ_{t+1} from GARCH, then r_{t+1} from σ_{t+1} , compute PnL, repeat 10k times.

Key: Model vol dynamics separately because options PnL is highly convex in vol."

2. How to test if one FX rate leads another? Does EUR/USD lead GBP/USD? **Answer:** "Granger causality test in VAR(1):

1. Run VAR on $(r_t^{\text{EUR/USD}}, r_t^{\text{GBP/USD}})$ with lags.
2. Test if EUR lags are jointly zero in GBP equation (F-test). If reject $\rightarrow$ EUR leads GBP.
3. Also check cross-correlation at lag 1: $\text{Corr}(r_t^{\text{EUR}}, r_{t-1}^{\text{GBP}})$.

16. Mini Exercises: Method Selection Scenarios

Exercise Method For each product, pick method + 2 pros + 1 con + variance reduction trick.

Exercise 1: 1-year American put

- **Method:** Binomial tree (100-200 steps)
- **Pro 1:** Handles early exercise exactly via backward induction
- **Pro 2:** Fast ($< 10\text{ms}$), accurate Delta/Gamma
- **Con:** Slow for Greeks vs BS formula (but who cares)
- **VR:** Not needed—tree is deterministic

Exercise 2: 3-month Asian call (average of 60 daily fixings)

- **Method:** MC with pathwise method for Greeks
- **Pro 1:** Path-dependent $\rightarrow$ MC natural
- **Pro 2:** Can use geometric Asian as control variate (exact price)
- **Con:** Slow with 60 steps (but daily fixings required)
- **VR:** Control variate $\rightarrow 5\times$ variance reduction

Exercise 3: Down-and-out call, barrier 90, spot 100, 6 months

- **Method:** PDE with non-uniform grid
- **Pro 1:** Fast, accurate near barrier
- **Pro 2:** Greeks at barrier available
- **Con:** Must align grid with barrier (dense at 90)
- **VR:** Brownian bridge for MC alternative, but PDE wins

Exercise 4: 5-asset basket European call

- **Method:** QMC (Sobol) with PCA factor reduction
- **Pro 1:** 5D $\rightarrow$ MC needed, PDE impossible
- **Pro 2:** PCA reduces effective dimension $\rightarrow$ QMC works well
- **Con:** Correlation matrix calibration tricky
- **VR:** Geometric basket as control variate

Exercise 5: Digital put, strike 100, 1 month

- **Method:** PDE with payoff smoothing
- **Pro 1:** PDE avoids MC variance explosion at strike
- **Pro 2:** Smooth payoff (e.g., call spread approximation) removes oscillations
- **Con:** Delta/Gamma spike at strike—hard to hedge
- **VR:** If MC, use importance sampling or likelihood ratio Greeks

Exercise 6: TARN (Target Accrual Redemption Note)

- **Method:** MC with early termination trigger
- **Pro 1:** Path-dependent knockout $\rightarrow$ MC only

Expected Positive Exposure (EPE):

$$\text{EPE} = \frac{1}{T} \int_0^T \text{EE}(t) dt$$

Potential Future Exposure (PFE):

$$\text{PFE}_\alpha(t) = \text{Quantile}_\alpha(E(t))$$

"Exposure = Positive MTM" **Memory:** Only count if you're winning. If MTM negative, exposure = 0 (they owe you, but you care about *their* default).

Exposure Profile for IRS:

Time (years)	EE	PFE (95%)
0	0	0
1	1% of notional	5%
5	3%	12%
10	2%	8%

Shape: Rises then falls (amortizing). Peak at 3-5 years for 10Y swap.

Exposure Drivers"Long maturity, high volatility, unilateral collateral → high exposure" $\text{EE}(t) = \max(V(t), 0) dP(t)$ "I compute exposure via MC: simulate rates to time t, mark swap, take $\max(0, \text{MTM})$. repeat across paths."

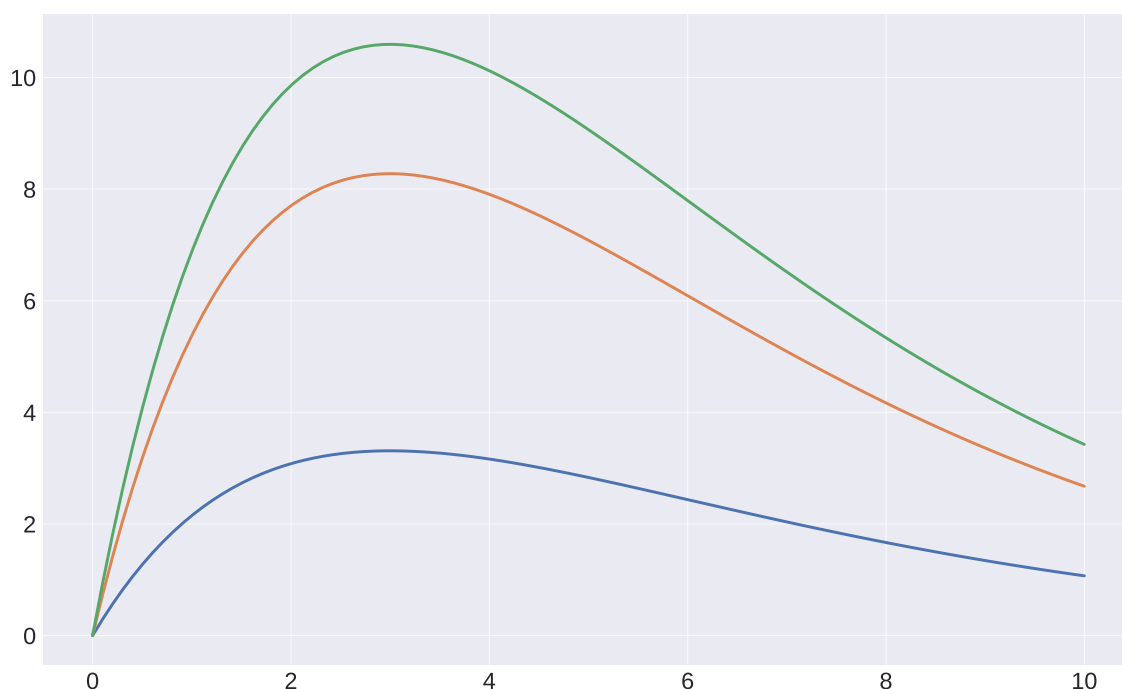


Figure 7: EE and PFE Profile for 10-Year Swap. Expected Exposure peaks mid-maturity. PFE shows tail exposure at 95% and 99% confidence.

9. Wrong-Way / Right-Way Risk: The Correlation Killer

Wrong-Way Risk (WWR):

$$\text{Corr}(E(t), \text{PD}_{\text{CP}}) > 0$$

Exposure increases as counterparty gets riskier.

Example: - Long CDS on Greece: If Greece defaults, your CDS pays out (exposure high) *and* counterparty (Greek bank) also likely defaults. **WWR = death.**